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Paleomagnetism and $^{40}Ar/^{39}Ar$-dating of Late Mesozoic Volcanic Pipes of Minusinsk Depression (Russia)

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Abstract. Volcanic pipes with K-Ar ages 65-49 Ma (Zubkov et.al., 1989) are widespread in the Minusinsk depression. We present the results of the first paleomagnetic investigation on four volcanic pipes (42 samples), accompanied by $^{40}Ar/^{39}Ar$ dating. All four pipes yield a high temperature ChRM component of reversed polarity with mean direction D=178.8, I=-62, n=33, k=160.7, a95=7.3. However, some samples from the Bele pipe demonstrate another direction (D=244.3, I=-58.2, n=4, k=407.4, a95=4.6). $^{40}Ar/^{39}Ar$ method gave plateau ages of pipes ranging from 74 to 81 Ma. These results cast some doubt on previous K-Ar dating. The paleomagnetic poles corresponding to both directions of magnetization are statistically different from the APWP of Eurasia given by Besse and Courtillot (1991). Paleomagnetic poles from basaltic rocks of the near-Baikal region (Khramov, 1984) are consistent with the first direction identified from Minusinsk depression, but again are different from the APWP of Eurasia. These data, therefore, call into question the validity of the Eurasian APWP to Northern Asia.
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1 Introduction

The Apparent Polar Wander Path (APWP) for Eurasia (Besse and Courtillot, 1991) is usually employed as a basic reference curve for Late-Mesozoic - Cenozoic paleomagnetic constrains of Asia (Thomas, et al, 1996, Kravchinsky et al, 1996, Zhao et al, 1996). Nevertheless, the majority of paleomagnetic data from volcanic rocks of Siberia (Khramov, 1984) are in a rather poor agreement with the Eurasian APWP. The question now arises of whether it is possible to extend the Eurasian APWP to Siberia. The only way to solve the problem is to perform paleomagnetic and $^{40}Ar/^{39}Ar$ investigations on volcanic rocks from within the Siberian Craton, which has been a stable tectonic regime since the Late Mesozoic. One of the most suitable areas for such investigations is the Minusinsk Depression, where volcanic pipes of Late Mesozoic - Early Cenozoic age intrude into slightly deformed Devonian and Carboniferous strata. In this paper we present the first paleomagnetic and radiometric results on four of these.

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2 Geological setting and sampling

2.1 Geology

The Minusinsk Depression is located within the Caledonian fold belt of Northern Asia (Fig. 1). This depression is of complicated tectonic origin, and its formation was accompanied by intensive alkaline-subalkaline Devonian volcanism. Brief reviews and some geodynamic interpretation of the development of this region are given by Zubkov et al. (1989) and Ashchepkov et al. (1995).

A ring of Mesozoic - Cenozoic alkaline-basalt volcanism has been found at the northern margin Minusinsk Depression. The volcanic pipes and dikes are grouped over the margins of Kopievo dome-shaped uplift and located near flexures and main faults (Fig.1). These volcanic bodies represent the root system of volcanic eruptions, forming volcanic pipes of complex structure, lava necks and numerous dikes. Pipes mostly are multiphase structures composed of eruptive breccias with altered groundmass containing abundant debris of basalts and hypabyssal basanites. No cover facies have been identified. Despite textural variety of the basalts, their chemical composition remains constant, and they are all classed as basanites. The volcanic rocks have been intruded into Devonian-Carboniferous age terrigenous deposits.

The K-Ar isotopic ages (Zubkov et al., 1989) for these four volcanic pipes are found within the range of 71-45 Ma. Two volcanoes (Bele and Tergeshskaya pipes),

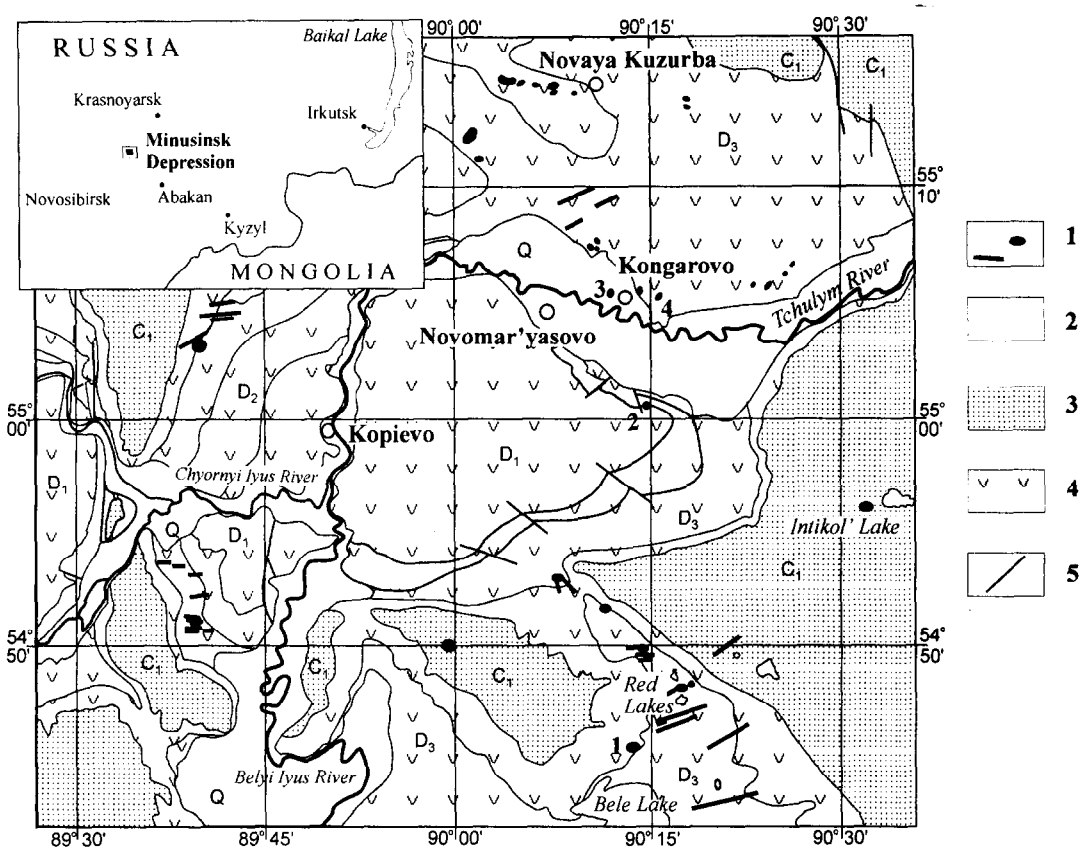


Fig.1. Geological sketch map of the northern part of the Minusinsk depression (redrawn from Geological map of the USSR 1:20000 scale). Symbols: 1- Late Mesosoic basaltic pipes and Devonian basaltic dykes; 2 - Quaternary deposits; 3 - Carboniferous sandstone and siltstone; 4 - Devonian volcanogenic-terrigenous deposits; 5 - faults. Volcanic pipes (in the map): 1 - Bele, 2 - Boradzhyl'skaya, 3 - Kongarovskaya, 4 - Setra.

containing garnet-bearing lherzolites, tend to be more ancient, giving K/Ar-ages of 62 and 65 Ma. The absolute isotopic dating performed for zircons from concentrates of the Bele pipe (dated in the Carnegie Institute, Geophysical Lab., USA) have yielded Pb^{206}/U^{238} ages 77.9 Ma (Ashchepkov et al., 1995).

2.2 Paleomagnetic sampling

42 paleomagnetic samples were collected from basalts of four pipes; the Bele, Sestra, Kongarovskaya and Boradzhyl'skaya pipes (Tab.1). Samples were collected as hand samples oriented by the magnetic compass with proper account of the influence of high-magnetic rocks. In the laboratory, the hand samples were cut up into standard cubic specimens by the size of a side of 20 mm.

3 Methods

All laboratory experiments were carried out in United Institute of Geology, Geophysics and Mineralogy Siberian Brunch of Russian Academy of Sciens using equipment of Paleomagnetic Laboratory and in Laboratory of Radiogenic and Stable Isotopes.

3.1 Paleomagnetism

Samples were measured using a JR-4 spinner-magnetometer. To isolate the characteristic component of magnetization (ChRM) the samples were subjected to stepwise thermal demagnetization up to 690°C (17 steps) in the Aparin 3-shielded furnace with an inner field of less than 10 nT. The magnetic vectors obtained during demagnetization were plotted in Zijderveld (1967) orthogonal projections. The computer program package designed by Enkin (1994) was applied for analysis and presentation of paleomagnetic data.

3.2 Ar/Ar-dating

Whole rock samples of basalt were investigated by $^{40}Ar/^{39}Ar$ method. The argon isotopic measurements were made on a M11201"B" mass spectrometer using a double vacuum Ta resistance furnace. Procedural blank (1500°C, 20 min) on ^{40}Ar was $1\text{-}2*10^{-13}$ mol. Age calculations were made using the plateau method applying the criteria of Feck et al. (1977).

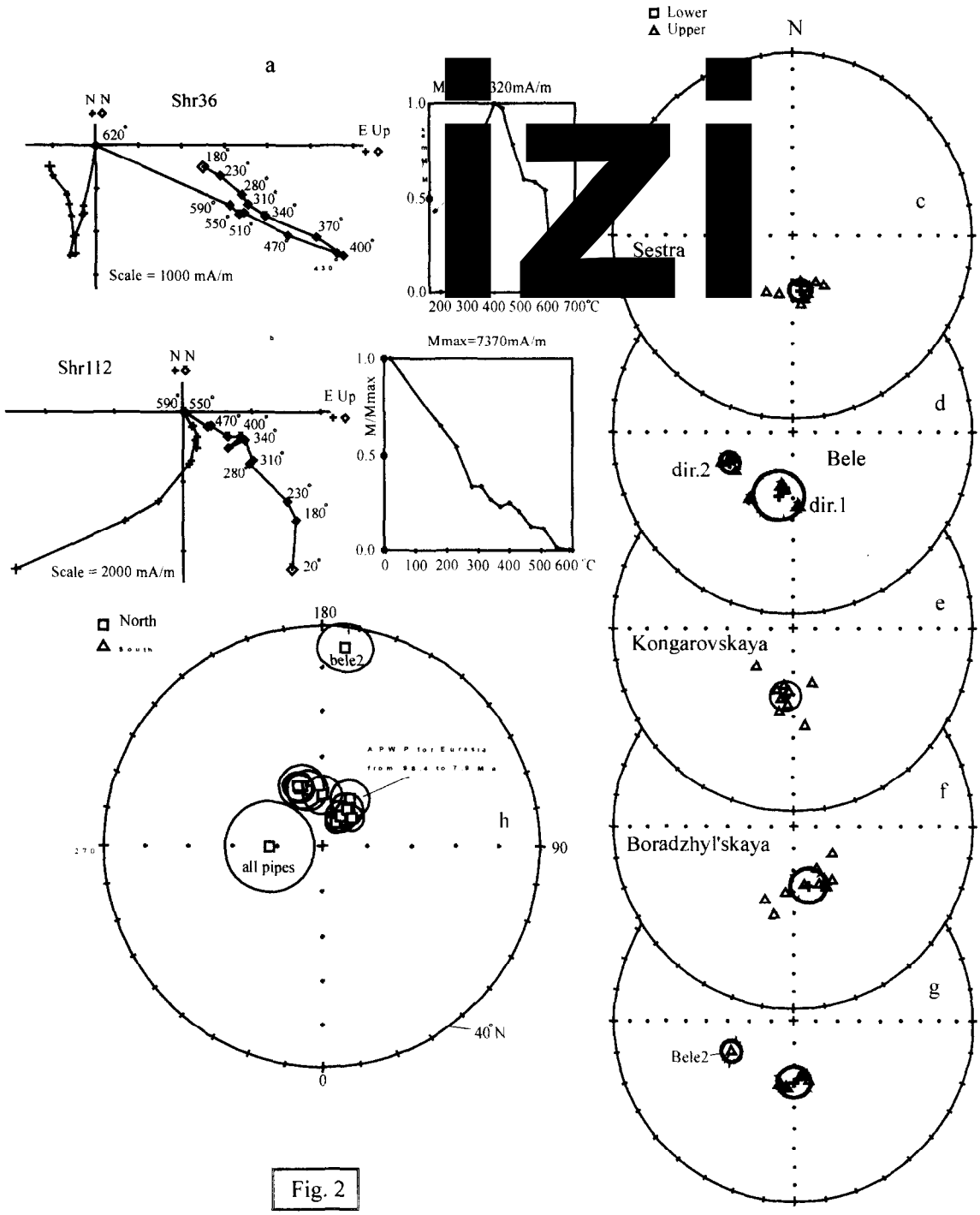


Fig. 2

Fig.2. Paleomagnetic results from volcanic pipes of Minusinsk depression. (a-b) Orthogonal vector diagrams and decay plots, that show the behavior of individual samples during thermal demagnetization (temperatures given in °C). (c-f) Equal area projections showing distribution of individual ChRM vectors and mean-pipe directions with corresponding cone of 95% confidence limits for pipes under study. (g) Equal area projection showing mean pipe direction and average direction for all pipes with corresponding confidence limits. Bele-2 direction is drawn separately. (h) Equal area projection showing pole positions and confidence limits calculated from average direction of this study and Bele-2 direction compared to Eurasian APWP (Besse and Courtillot, 1991).

Table 1. Paleomagnetic data from volcanic pipes of the Minusinsk depression (Russia)

				Mean direction					VGP		(normal polarity)	
Pipe	Lt N (°)	Lg E (°)	n/N	Dec (°)	Inc (°)	K	a95 (°)	Plat. (°)	Lat. (°)	Lon. (°)	dp (°)	dm (°)
Boradzhyl'skaya	55	90.25	10/10	165.7	-62	37.3	8	-43.4	75.1	314.7	9.7	12.4
Bele	54.75	90.25	4/13	192.7	-60.5	68.2	11.2	-41.5	74.3	232.8	13	17
Kongarovskaya	55.1	90.2	9/9	186.6	-59.1	58.2	6.8	-39.9	74.2	251.4	7.6	10.2
Sestra	55.1	90.27	10/10	172.2	-64.7	97	4.9	-46.6	80.2	303.5	6.3	7.9
Mean for 4 pipes			4	179.8	-62	160.7	7.3	-43.2	78.3	271	8.8	11.3
Bele (direction 2)	54.75	90.25	4/13	244.3	-58.2	407.4	4.6	-38.9	45	173.2	5	6.8

4 Results

4.1 Paleomagnetism

Two directions of magnetization can be resolved by thermal demagnetization (Fig.2). The first direction is of normal polarity, and is usually removed bellow 400°C. It has a large scatter and most likely represents viscous magnetization or its remains. The high temperature component of magnetization, defined by univectoral decay towards the origin, is isolated generally between 450°C and 600°C. This direction is termed the ChRM and is identified only with reversed polarity. The sample ChRM directions group tightly into compact clusters with mean-pipe directions in close agreement. Only four samples from the Bele pipe show anomalous directions (Bele-2) different from the other pipes. The overall mean characteristic direction (excluding Bele-2 directions) were computed from four mean directions for different pipes (Fig.2 and Tab. 1)

4.2 Ar/Ar dating

Typical age spectra (Bele and Boradzhyl'skaya pipes) are shown in Fig. 3. They produce well-defined 3-4 step plateaus with ages of 79±2 Ma and 81±1.2 Ma accordingly. The $^{40}Ar/^{39}Ar$ plateau ages and K-Ar data are given in the Tab.2. It must be mentioned that sample BG-1 represents the host for sanidine megacryst BG-671 and their absolute ages are in a good agreement.

Table 2. Summary of $^{40}Ar/^{39}Ar$ results from volcanic pipes of the Minusinsk depression (Russia) in comparison with K-Ar and $^{206}Pb/^{238}U$ isotopic ages

Pipe	$^{40}Ar/^{39}Ar$ data		K-Ar-age,	$^{206}Pb/^{238}U$-age,
	Sample	Age, Ma	Ma[2]	Ma[3]
Bele	Bele	79±2.0	62±3	77.9
Kongarovskaya	KG-2	74±5.5	45±2	
	BG-D	81±1.2	49±3	
Boradzhyl'skaya	BG-1	77±2.1	53±3	
	BG-671[1]	77±5.0		

1) sanidine megacryst, 2) Zubkov et al. (1989), 3) zircon heavy concentrate (Ashchepkov et al., 1995)

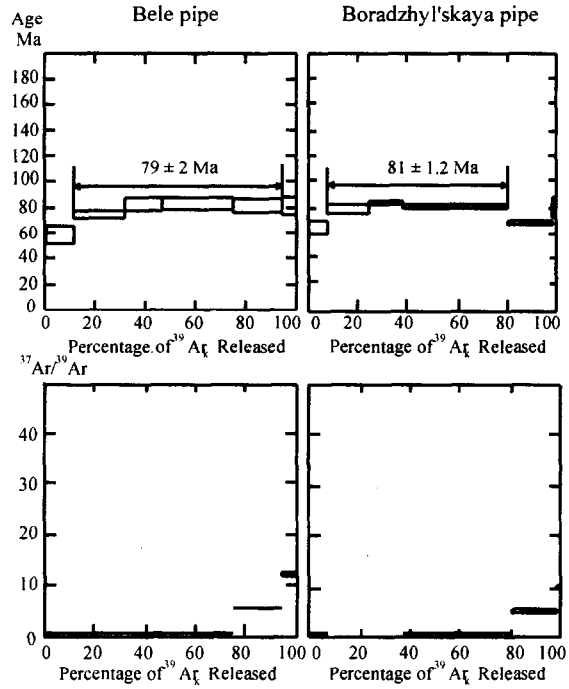


Fig.3. $^{40}Ar/^{39}Ar$ apparent age spectra for Bele and Boradzhyl'skaya pipes. The vertical width of each bar represents the analytical error in the age of the individual temperature step at ±1σ.

5 Discussion

Disparity between $^{40}Ar/^{39}Ar$ and K-Ar data are probably caused by samples character. The inner structure of explosive pipe is sufficiently heterogeneous to carry out whole rock measurements. However, minerals here are vulnerable to surface factors and secondary thermal processes. Using step heating in Ar-Ar measurements it is possible divide gas fractions from secondary changed and fresh minerals. In this case the Ar-Ar plateau age seems to be more closely to the rock origin time than K-Ar age. The K-Ar dating based on the consideration of all sample gas quantity the apparent K-Ar age may be lowered.

Reversed polarity of ChRM directions for all pipes well correlate with obtained $^{40}Ar/^{39}Ar$ ages (taking into account confidence limits) and probably represents the end of 33R reversed polarity chron of the Magnetic Polarity Time Scale (Harland et al., 1989), whereas variations in K-Ar ages imply the presence both normal and reversed polarity for pipes with different ages.

Distinctions between paleomagnetic poles obtained and poles of Eurasian APWP (Besse and Courtillot, 1991) are statistically significant as their confidence limits do not overlap (Fig. 2). Furthermore most of the Cretaceous and Cenozoic paleomagnetic directions obtained from the neighbor Baikal region (Khramov, 1984) and Gorny Altai (Thomas et al., 1996) also exhibit counterclockwise rotation relative to «stable Asia». The similar trend in discordance suggests that it is not random and may reflect real motions of Siberian blocks.

6 Conclusions

Ar/Ar dates from volcanic pipes from Minusinsk depression yield apparent ages of about 80 Ma, which we consider to be the best estimation for investigated pipes. Paleomagnetic data from the pipes yield well-defined reversed polarity magnetization with a virtual geomagnetic pole at 78.3°N, 271°E (A95=9.9°). Our $^{40}Ar/^{39}Ar$ dates indicates that the magnetization was probably acquired at the end of chron 33R (Harland et al., 1989). The virtual geomagnetic pole obtained in this study statistically differs from the reference APWP for Eurasia of Besse and Courtillot (1991) and thus calls into the question the implementation of Eurasian APWP to Northern Asia.

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